Advanced Color Theory in Relations to Primes

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Abstract

We equip the admissible residue shell $S = \{1, 2, 4, 5, 7, 8\} \subset \mathbb{Z}/9\mathbb{Z}$ with a finitely generated chromatic algebra. The construction incorporates the detailed mixing rules and complementary structure recorded in the co-authors' handwritten notes and remains compatible with the modular background of the twin-pair residue theorem and the cubic residue classification.

1 Introduction and Modular Background

Advanced Color Theory is defined on the residue classes occupied by primes greater than 3. As modular background, the twin-pair residue theorem asserts that every twin-prime pair $p, p+2 > 5$ realizes one of $(2, 4)$, $(5, 7)$, or $(8, 1)$ in $\mathbb{Z}/9\mathbb{Z}$, while every prime $p > 3$ satisfies $p \bmod 9 \in S$. Independently,

$$n^3 \equiv 0, 1, 8 \pmod 9 \qquad (n \in \mathbb{Z}), \tag{1}$$

so that three-cube sums exclude classes 4 and 5 while four-cube sums are surjective. These facts delimit the domain.

2 Primary Channel Operators

Four primary generators act as independent channel operators:

$$R \leftrightarrow 3 \qquad \text{(hue-temperature increment } \partial_T^+), \tag{2}$$

$$B \leftrightarrow 7 \qquad \text{(hue-temperature decrement } \partial_T^-), \tag{3}$$

$$W \leftrightarrow 1 \qquad \text{(value increment } \partial_V^+), \tag{4}$$

$$\mathrm{Bl} \leftrightarrow 9 \qquad \text{(value decrement } \partial_V^-). \tag{5}$$

Complementary cancellation:

$$R + B = 0_T, \qquad W + \mathrm{Bl} = 0_V. \tag{6}$$

3 Secondary Elements, Mixing Rules, and Complementary Structure

Secondary elements arise by simultaneous channel adjustment. The handwritten notes establish:

$$6 \leftrightarrow R + W \qquad \text{(warm + bright)}, \tag{7}$$

$$4 \leftrightarrow B + \mathrm{Bl} \qquad \text{(cool + dark)}, \tag{8}$$

$$2 \leftrightarrow B + W \qquad \text{(cool + bright)}, \tag{9}$$

$$8 \leftrightarrow R + \mathrm{Bl} \qquad \text{(warm + dark)}. \tag{10}$$

Updated complementarity: Magenta complementary to Green; Cyan complementary to Yellow. Explicit cancellation rules include Red + Blue (temperature), White + Black (value), and the composite pairs Magenta + Green, Cyan + Yellow. Secondary mixtures adjust both channels; tertiary mixtures produce smaller shifts emphasizing primary colors. The resulting set coincides with S, establishing algebraic closure.

4 Induced Coloring and Mixture Map

The coloring $c : S \to \{\text{chromatic elements}\}$ is the identity on the labels. For $p > 3$ set $c(p) := c(p \bmod 9)$. For $(a, b) \in S \times S$ the mixture is

$$m(a, b) = c(a) + c(b) \tag{11}$$

reduced by the complementary relations.

5 Explicit Mixture Examples

Residue pair	Classes	Ingredients	Reduced mixture
(11,13)	(2,4)	Cyan + Green	cool-dark composite
(17,19)	(8,1)	Magenta + White	warm-bright composite
(41,43)	(5,7)	Blue + Blue	pure cool

6 Higher-Order Terms and Compatibility

Any finite sum of secondary elements reduces to an integer linear combination of the four primaries whose support remains inside S. Complementary cancellation operates at every order. Compatibility with the cubic residue classification is immediate.

7 Conclusions

The chromatic algebra is closed on S, the induced coloring of primes greater than 3 is well-defined, and the construction is compatible with the modular background. The mixing rules are fully determined by the residue arithmetic together with the channel-operator assignments.

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References

[1] Johnson, E. EJ's Alternate Base Prime Number Pattern and the Twin Prime Counterflow Model. Zenodo, 2026.

[2] Neer, S. Modular analyses of cubic residues, 2026.

[3] Johnson, E. Handwritten chromatic mixing rules and complementary structure, 2026.